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QEST HACK 2026

PARTNER CHALLENGE

Joint optimisation of a hybrid portfolio

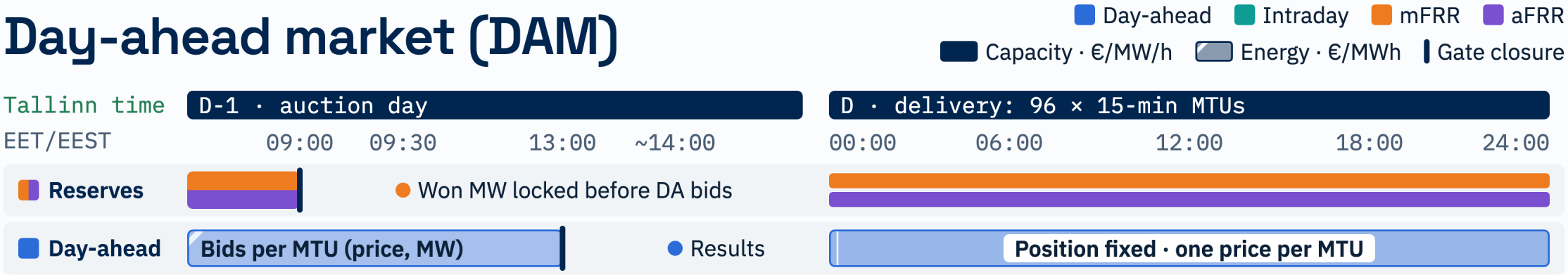
Batteries, solar and wind parks and demand response are managed as one portfolio. For every hour, we decide which asset to offer to which market, under weather-forecast uncertainty and imbalance costs.

The slides that follow set out the market rules any solution has to respect, then the toy problem, how it is scored, and the dataset.

IN THIS DECK

01 Day-ahead · 02 Pricing · 03 Intraday · 04 mFRR · 05 aFRR · 06 Overview · 07 Imbalance · 08 Hybrid sites · 09 Hindsight or foresight · 10 Assets · 11 Portfolio puzzle · 12 Scoring · 13 Dataset

Day-ahead market (DAM)



HOW IT WORKS

01 One daily auction: the European SDAC, traded via Nord Pool.

02 One clearing price per 15-min MTU, per bidding zone (EE, LV, LT).

03 Gate closure D-1 13:00 (12:00 CET), results ~14:00. The position is then fixed for all 96 MTUs of D.

Room for day-ahead bids, per direction



Won reserves first · power and SoC limit = full bar

RULES & LIMITATIONS

A binding position for the whole delivery day

- Binding: deviations from the position are settled as imbalance.
- Bids must fit power and state-of-charge limits after subtracting balancing capacity already won.
- Minimum volume step: 0.1 MW.

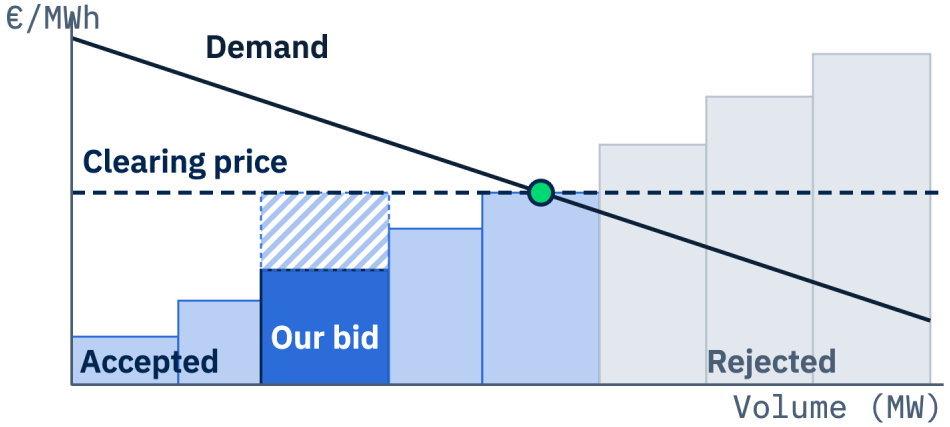
Times in Tallinn local time. MTU = 15 min.

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How prices are cleared

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

DAY-AHEAD CLEARING · ILLUSTRATIVE



Our bid sits below the clearing price, so it is accepted and paid the clearing price (hatched gap), not its own bid.

HOW EACH PRICE IS SET

Day-ahead	One European auction sets a single price per 15 min for each country.
mFRR capacity	One published price per 15 min and direction: treat it as what an accepted bid earns.
mFRR energy	One price per 15 min, set by the last activated bid. UP and DOWN both settle at it.
aFRR capacity	Same as mFRR capacity.
aFRR energy	The price changes every 4 seconds; the data has every tick.

Bid price ≠ paid price
Except intraday, the bid decides acceptance, not the price.

Two payments, one MTU
Capacity and energy can both pay on the same MTU.

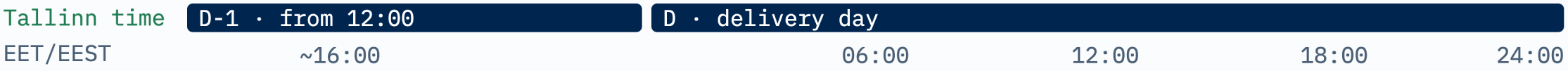
Energy prices swing
Balancing energy prices move far more than day-ahead.

Times in Tallinn local time. MTU = 15 min.

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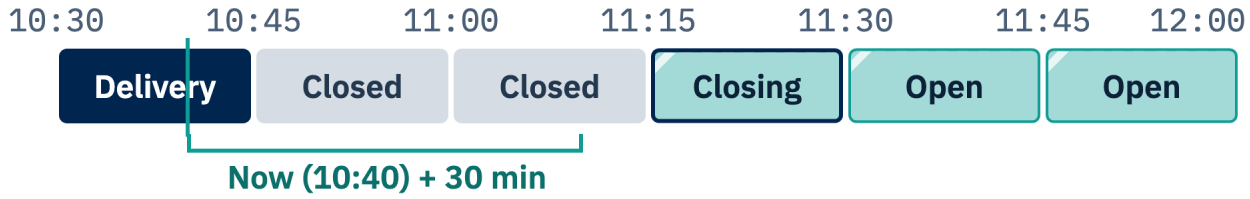
Intraday market (IDM)

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure



Continuous — Opens D-1 afternoon · trade any MTU of D until its T–30

SLIDING WINDOW · EXAMPLE AT 10:40 ON D



Each MTU closes 30 min before it starts. At 10:45 the 11:15 MTU closes, and the window slides one MTU on every 15 min.

USE IT TO — Adjust the DA position · React to forecasts · Re-optimise SoC

RULES & LIMITATIONS

Continuous, pay-as-bid

- Each trade is matched and priced separately.
- Gate closure T–30 for every MTU.
- Same power, SoC and reserved-capacity limits as day-ahead.

Toy problem: no intraday, so the day-ahead position is final.

Times in Tallinn local time. MTU = 15 min.

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mFRR: capacity + energy

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

Tallinn time	D-1 · capacity auction			D · every MTU · manual, within minutes		
EET/EEST	09:00	~09:30		T−25	T	T+15
Capacity €/MW/h	**Bids · UP / DOWN**	Results	**Won MW held**	**Available in every won MTU, per direction**		
Energy €/MWh		*Then bid energy for every won MTU →*		**Energy bids ≥ won MW**		**Activation**

RULES & LIMITATIONS

Capacity = obligation
Each won MTU and direction needs an energy bid ≥ the won MW before T−25.

Gate closure T−25
Energy bids close 25 min before delivery. No changes or withdrawals after that.

1 MW steps
Reserve bids in whole MW only, minimum 1 MW.

Which bids are activated
UP if price ≥ bid, DOWN if price ≤ bid.

Headroom cap
mFRR + aFRR per direction ≤ physical headroom: grid, power and SoC.

Backed by SoC
Won MW needs the energy to sustain activation, so it can't be used in day-ahead or intraday.

Times in Tallinn local time. MTU = 15 min.

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aFRR: capacity + energy

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

Tallinn time	D-1 · capacity auction			D · every MTU · automatic, in seconds		
EET/EEST	09:00	~09:30		T−25	T	T+15
Capacity €/MW/h	**Bids · UP / DOWN**	Results	**Won MW held**	**Available in every won MTU, per direction**		
Energy €/MWh	*Then bid energy for every won MTU →*			**Energy bids ≥ won MW**		**Automatic**

RULES & LIMITATIONS

Capacity = obligation
As with mFRR, each won MTU and direction needs an energy bid ≥ the won MW.

1 MW steps · T−25
Whole MW only, minimum 1 MW. Energy bids close at T−25.

SoC room both ways
Follows the TSO signal every few seconds, UP and DOWN. Keep SoC room in both directions.

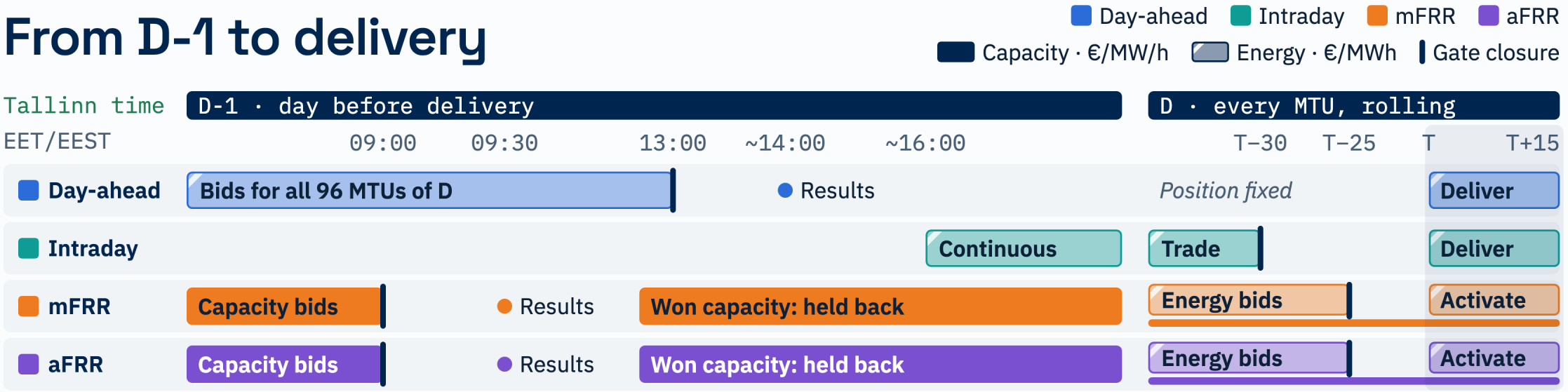
Partial activation
The signal moves every few seconds, so an MTU's activated energy is usually only part of the bid.

KratTrade today: aFRR before mFRR
When headroom is shared, we cover aFRR first. A baseline to beat, not a rule.

Times in Tallinn local time. MTU = 15 min.

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From D-1 to delivery



One battery's MW, filled bottom-up



KratTrade today: aFRR → mFRR

Bids that back won capacity use its MW.

RULES & LIMITATIONS

Capacity is decided first. Energy markets fill the remaining room.

- The capacity auction closes D-1 09:00, before day-ahead (13:00).
- Capacity not won stays free for day-ahead and intraday.
- Won capacity must stay available: its MW can't be used in day-ahead or intraday, per MTU and direction.
- Reserves are bid per direction: UP (discharge) and DOWN (charge).

Times in Tallinn local time. MTU = 15 min.

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Imbalance settlement

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

Tallinn time	D · delivery	Rest of month	Following month
EET/EEST	T T+15		4th · 05:30

Delivery — **Metered MTU** — Settled per 15-min MTU, per bidding zone

Price — Preliminary · near real time — **Final price published**

HOW IT'S APPLIED

Committed = day-ahead + intraday + activated UP − activated DOWN (intraday = 0 in the toy problem)

Imbalance = metered − committed, settled at imbalance × price

Worked example · one MTU · illustrative

DA +2.0 + ID −0.5 + UP +0.5 − DOWN 0.0 **= 2.0 MWh committed**

Metered 1.8 − committed 2.0 = **−0.2 MWh** (short)

−0.2 MWh × €120/MWh = **−€24**: we pay €24

SIGN CONVENTION

+ = power into the grid: export, discharge, produce.

− = power out of the grid: import, charge.

Imbalance × price above 0: we are paid. Below 0: we pay.

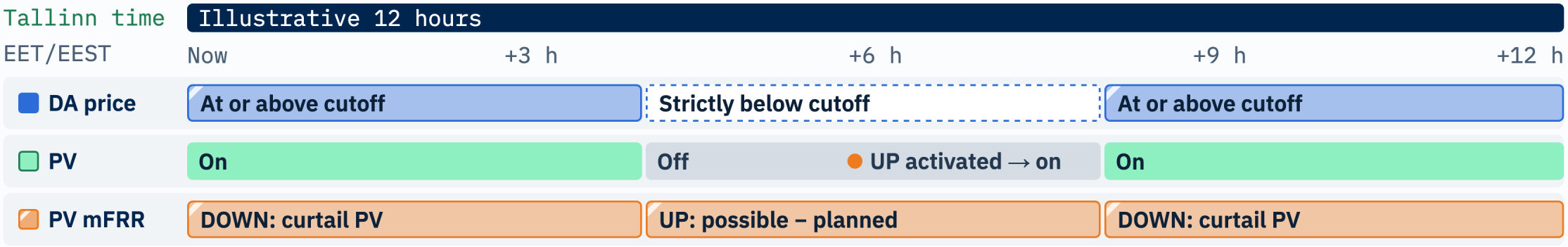
Toy problem: only needed at P4 (prospective) on the hybrid site, where PV forecast misses settle as imbalance.

Times in Tallinn local time. MTU = 15 min.

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Hybrid PV + battery

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure



RULES & LIMITATIONS

One grid connection

PV and battery share it: their combined flow stays within the connection limit.

No double-counting

Battery charging can't count on PV output for import room: PV can drop within the MTU.

PV price cutoff

Below PV's break-even price, e.g. a negative DA price, producing loses money: curtail.

PV in mFRR

Running PV can offer DOWN by curtailing. Curtailed PV can offer UP.

KratTrade today: PV before battery

In mFRR we offer PV volume before the battery's. A baseline to beat, not a rule.

Times in Tallinn local time. MTU = 15 min.

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Hindsight or foresight

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

RETROSPECTIVE · HINDSIGHT

Prices are known. What should we have bid?

- A deterministic optimisation: it maps straight onto a QUBO (binary quadratic model).
- Classical solvers find the true optimum, so the quantum answer has a measurable gap.
- No forecasts and no imbalance: delivery follows the plan.
- No financial cycles: no DA buy with an UP bid, or DA sell with a DOWN bid, in one MTU.

PROSPECTIVE · FORESIGHT

Prices are unknown. Bid before each gate.

- Gates: capacity D-1 09:00, day-ahead D-1 13:00, energy bids T−25.
- Needs price and PV forecasts: mostly classical machine learning.
- No future data. Forecast misses settle as imbalance (slide 7).
- Optimising over several price scenarios multiplies the binaries.
- Financial cycles allowed, if each leg can be delivered on its own.

WHERE TO START

Start retrospective (P1 to P3): it isolates the optimisation, where quantum applies. Prospective is the last part of the stretch goal, P4.

Times in Tallinn local time. MTU = 15 min.

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The two asset setups

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

BATTERY

100 MW
power, charge or discharge

200 MWh
energy: 2 hours at full power

Grid connection: not binding.

HYBRID SITE

20 MW
solar PV

20 MW
battery, 40 MWh

20 MW
grid, import and export

PV: the Solarpark forecast scaled to 20 MW (README).

Battery limits

Position plus reserved MW within ±100 MW. SoC within 0–200 MWh, with energy behind every reserve.

Hybrid limits

PV + discharge − charge within ±20 MW at the meter. At full PV output the battery can only charge, unless PV is curtailed.

Both setups: 100% round-trip efficiency, state of charge 0–100%, no cycling cost.

Grid tariff on every MWh imported: €30/MWh Mon–Fri 07:00–22:00 Estonian time, €20/MWh at night, weekends and holidays. Exports pay none.

Times in Tallinn local time. MTU = 15 min.

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The portfolio puzzle

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

TWO SITES · ONE HOUR

Battery 100 MW / 200 MWh

Hybrid site: PV + battery

PV 20 MW · battery 20 MW / 40 MWh · one 20 MW connection

BATTERY, HYBRID BATTERY AND HYBRID PV EACH PICK ONE

DA sell | DA buy | idle | mFRR UP | mFRR DOWN

Dashed options come in at P2. Prices are known: maximise €.

RULES

- Hybrid: PV + battery together within the 20 MW connection.
- Batteries start at 50% SoC and sell only what they hold; leftover energy is worth the day's average DA price.
- Imports pay the grid tariff (slide 10).
- PV can't exceed its forecast. No financial cycles.

WHY IT SUITS QUANTUM

- P2: 3 decisions × 5 options ≈ 15 qubits per hour.
- P2: at most $5^3 = 125$ combinations: a laptop checks them all.
- Same sites, whole day: 5^288 ≈ 10^201 combinations.

PORTFOLIO LADDER

P1 · 9 qubits
1 hour, day-ahead only

P2 · 15 qubits
+ mFRR UP and DOWN

P3 · 30 qubits
+ a 2nd hour, batteries linked by SoC

P4 · stretch
aFRR → capacity → stepped MW → prospective

Times in Tallinn local time. MTU = 15 min.

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Scoring the puzzle

Day-ahead · Intraday · mFRR · aFRR
Capacity · €/MW/h · Energy · €/MWh · Gate closure

SAME € RULES FROM P1 TO P4

Score (€) = day-ahead + capacity + reserve energy + energy left − grid tariff − imbalance

No future data (prospective): each decision uses only what was published before its gate closure.

Compare: retrospective → gap to the classical optimum; prospective → € on days not in the data.

Day-ahead

Position (MW) × 0.25 h × day-ahead price. Selling earns, buying costs. Energy left at the end is worth the day's average DA price.

Capacity

Every capacity bid counts as accepted at the published price: MW × €/MW/h × 0.25 h per MTU.

mFRR energy

In the money (UP: price ≥ bid, DOWN: price ≤ bid)
The whole bid counts as activated for the MTU, at the mFRR price.

aFRR energy

Use 4-s PICASSO prices. Each in-the-money tick (UP: price ≥ bid, DOWN: price ≤ bid) activates the bid: MW × 4 s at that tick's price.

Imbalance

Prospective only: (metered − committed) × imbalance price, as on slide 7. Hybrid: the latest forecast stands in for metered PV.

Who pays for energy

UP: you are paid price × MWh. DOWN: you pay it, so a negative DOWN price pays you. Imports also pay the grid tariff.

Times in Tallinn local time. MTU = 15 min.

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The dataset

Estonia (EE) · delivery days 1 Jan – 25 Oct 2026

3 Parquet files + README

15 MIN · 28,608 ROWS

prices_15min

Day-ahead, mFRR, imbalance and capacity prices

15 MIN · 28,608 ROWS

solarpark_forecast_15min

42.3 MW solar park: latest and day-before forecast, MW

4 S · 6.4 M ROWS

afrr_picasso_4s

The only aFRR prices: every 4 s, as published. Aggregate them yourself.

Join on UTC time

datetime_utc is the start of each interval. datetime_local is Tallinn time.

Use the D-1 forecast

For day-ahead bids, use solarpark_forecast_d1_mw. The latest forecast knows more.

Empty is not zero

An empty aFRR price means no price in that direction. Nothing is gap-filled.

Sources: Nord Pool · Baltic Transparency Dashboard · PICASSO

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